# Metaheuristics & combinatorial optimisation Introduction

dr. Stefaan Haspeslagh

# Who am I?

- Senior researcher @ KU Leuven – PhD in computer science in 2012.
  - Background in metaheuristic, exact and hybrid algorithms to tackle combinatorial optimisation problems (planning, scheduling, cutting, packing, vehicle routing, group composition, …)

- Innovation manager AI @ Vives University of applied sciences
  - Applications in industry, biomedical sciences, health, transport, logistics, workforce planning, …
  - AI Adoption!

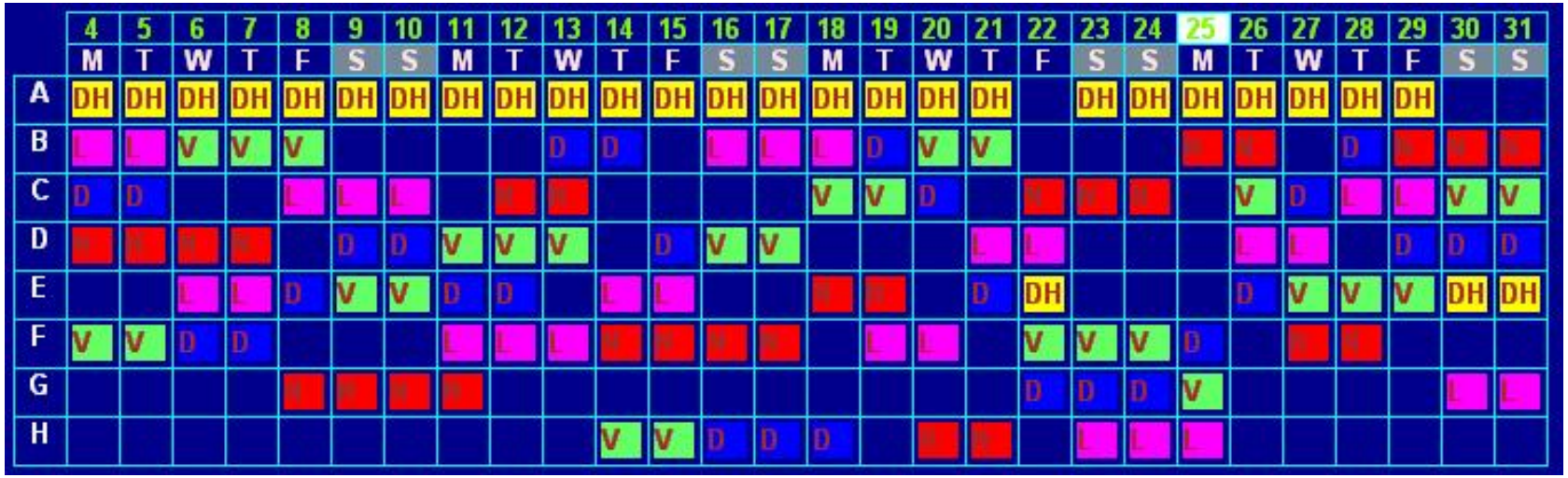
- Teaching:
  - Lecturer in AI@Vives: optimisation – machine learning – deep learning
  - Visiting professor @ KU Leuven: combinatorial optimization and metaheuristics in advanced master in AI for business and industry

# Combinatorial optimisation?

- Applications
- What?

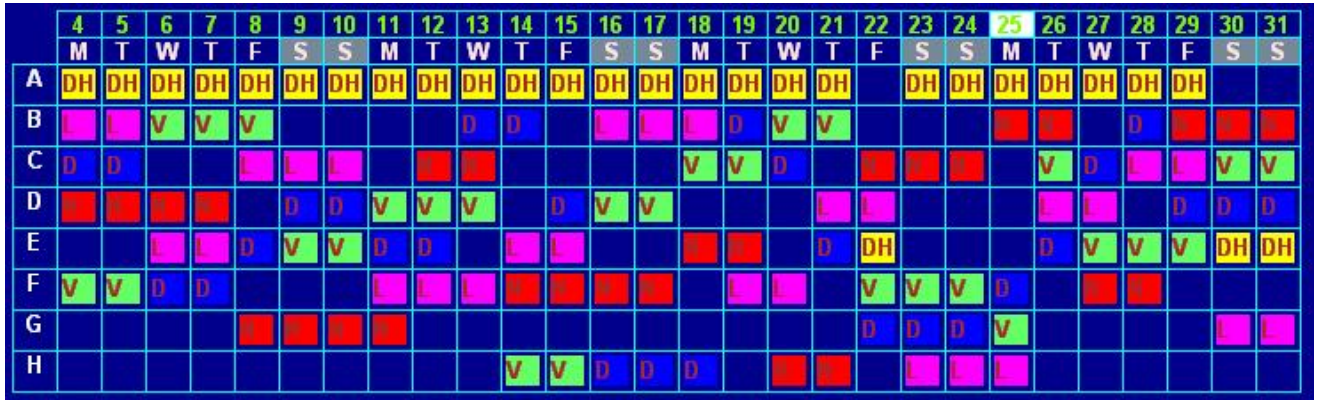
# AI and planning

- Problem:
  - assignments of shifts to employees with a certain qualification
  - considering a large number of constraints and preferences



# AI and planning

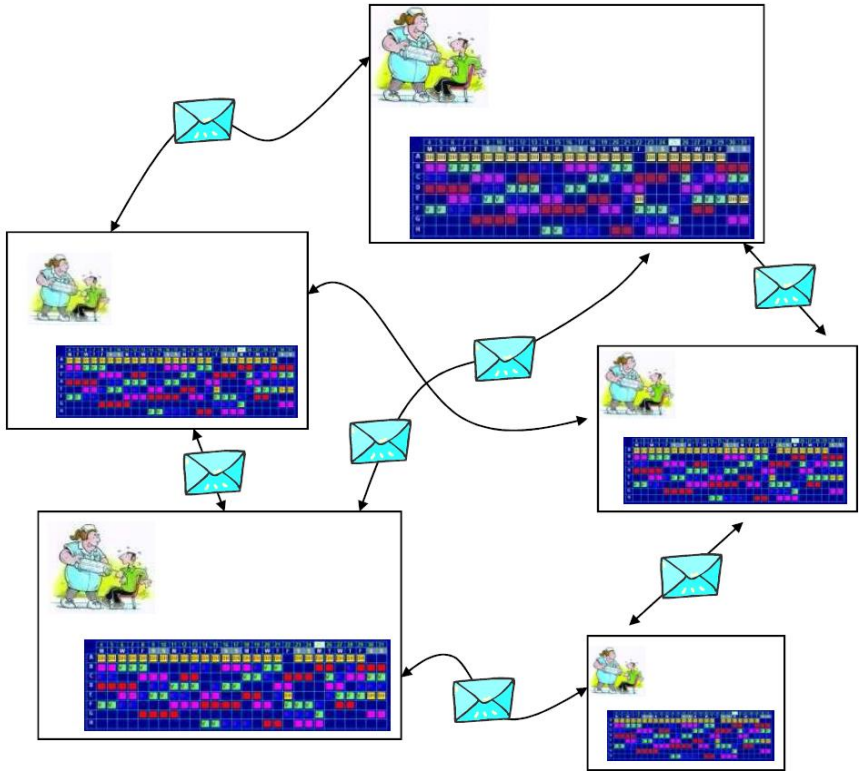
- **<u>Problem size:</u>**
  - Small example:
    - 4 shifts to schedule
    - 10 possible employees
    - 7 days

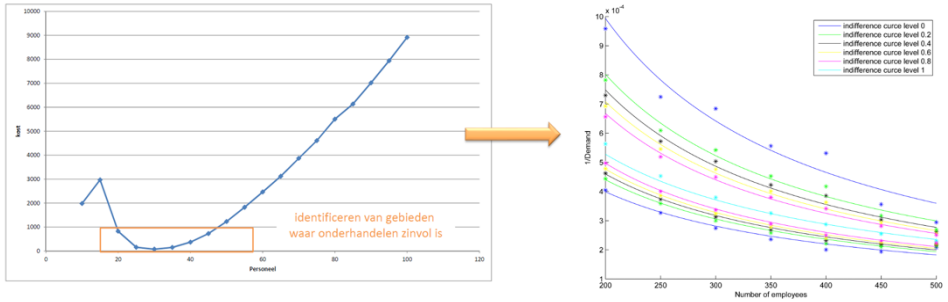


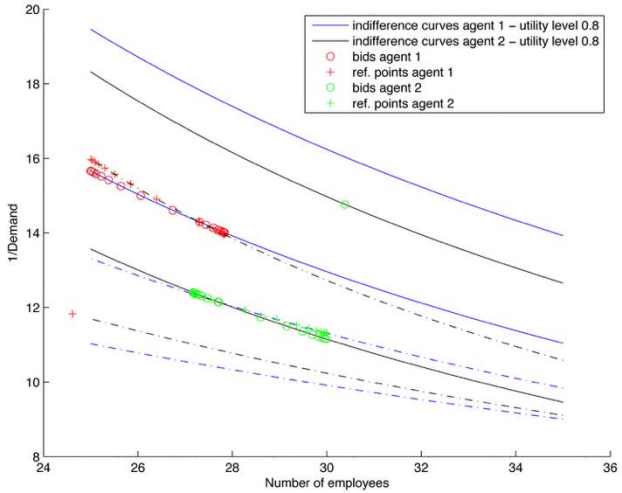

  - Number of variations on 1 day = 5040
    - For 1 week: number of possible rosters = $5040^7$ = 8,26**e**25
    - For 1 month: number of possible rosters = $5040^{30}$ = 1,18**e**111

# AI and planning

- Workforce planning as an example – resource planning

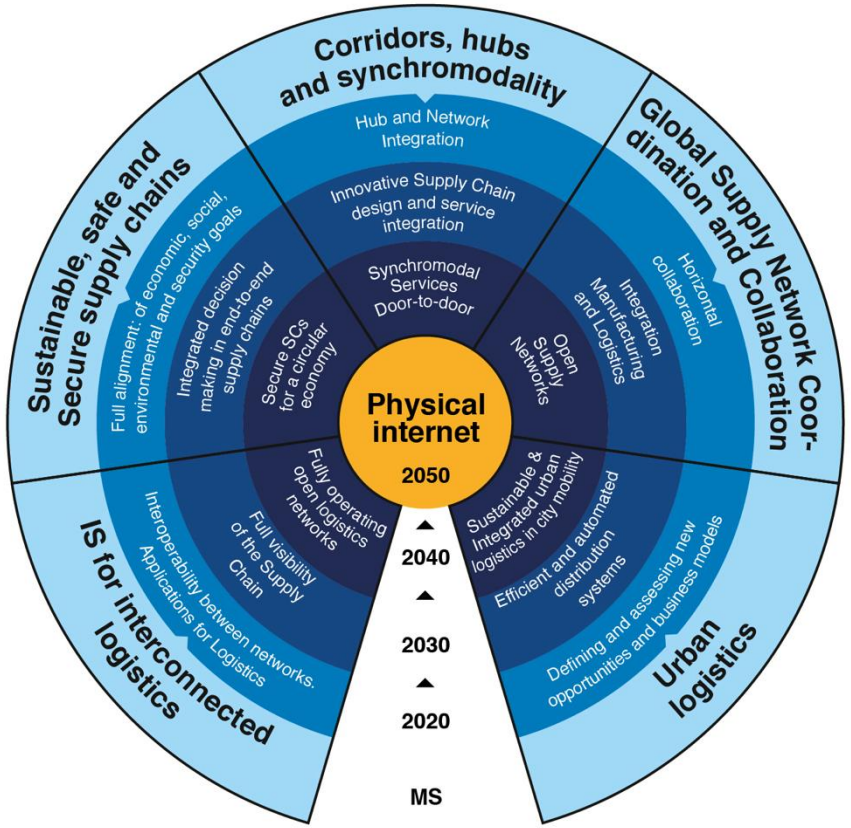


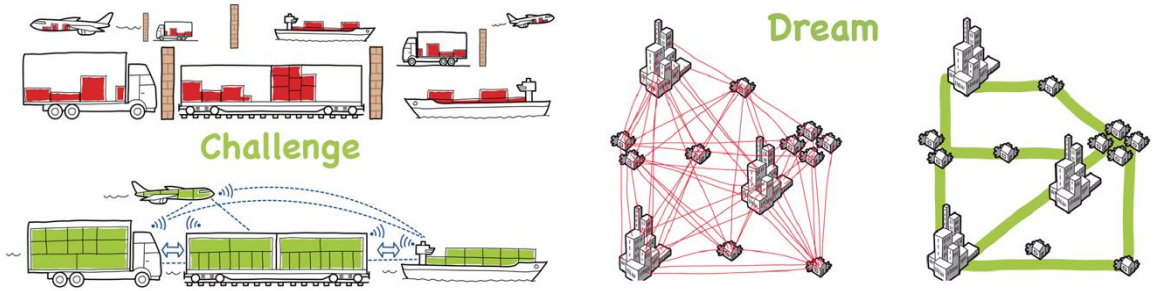


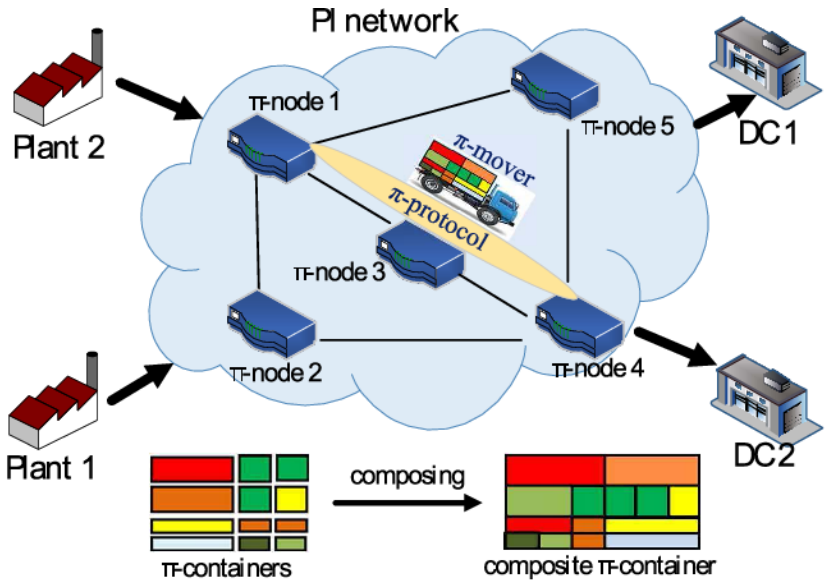



# AI – synchromodal transport

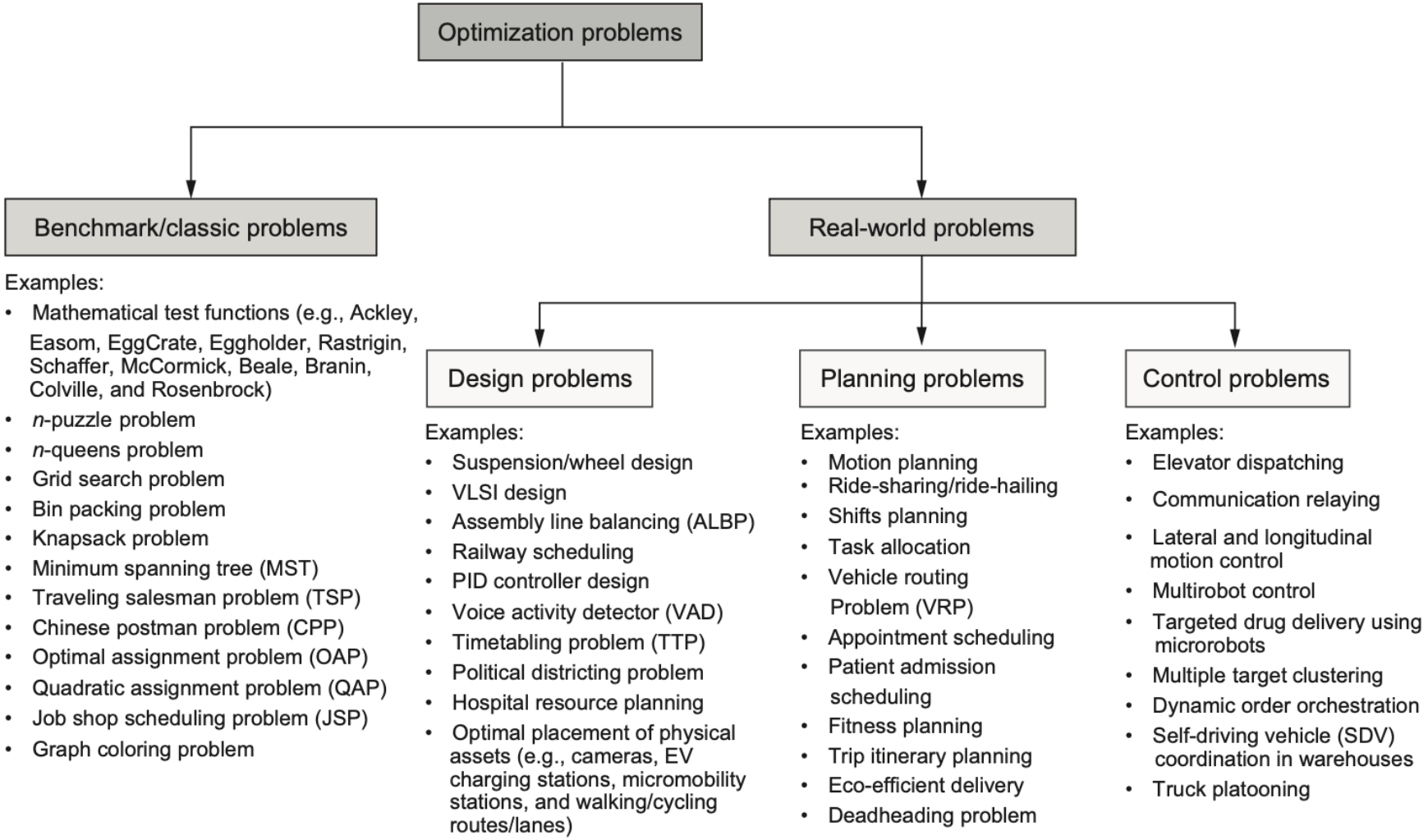
- Physical internet



# Real world examples



# Optimisation

- Optimisation refers to the practice of finding the “best” solutions to a given problem, where “best” usually means satisfactory or acceptable, possibly subject to a given set of constraints.

- E.g. finding the “best” configuration or set of parameters

# Optimisation

- General definition:
  - Find x to

$$\begin{aligned} &minimise\ f(x) \\ &subject\ to\ g_i(x) \geq 0. \quad \text{i=1,...,m} \\ &\qquad\qquad h_j(x) = 0 \quad \text{j=1,…,p} \end{aligned}$$

- Special types:
  - *f* convex, $g_i$ concave, $h_j$ linear → convex programming problem
  - if all linear → linear programming problem → solution is among a finite set of possible solutions → combinatorial problem

# Optimisation problems

- Two categories: continuous and discrete variables
- Discrete variables → combinatorial problems
- Solution is typically an object from a finite, or possibly countably infinite, set — typically an integer, set, permutation, or graph.

# Optimisation problems

- Definition:
  - An instance of an optimisation problem is a pair (F, c) where F is any set, the domain of feasible points (solutions), c is the cost function

$$c : F \to \mathbb{R}$$

  - The problem is to find an $\mathrm{f} \in \mathrm{F}$ for which:

$$c(f) \leq c(y) \text{ for all } y \in F$$

# Optimisation problems

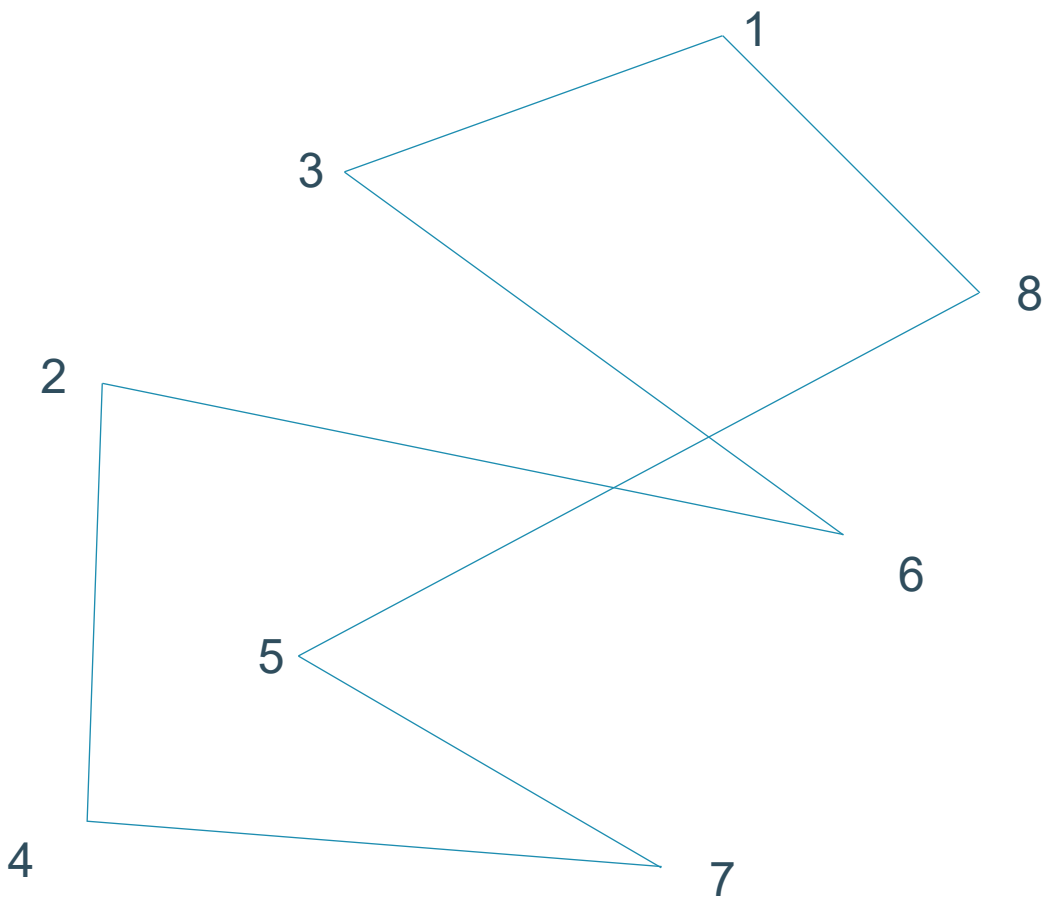
- Definition:
  - An optimisation problem is a set I of instances of an optimisation problem.

# Optimisation problems

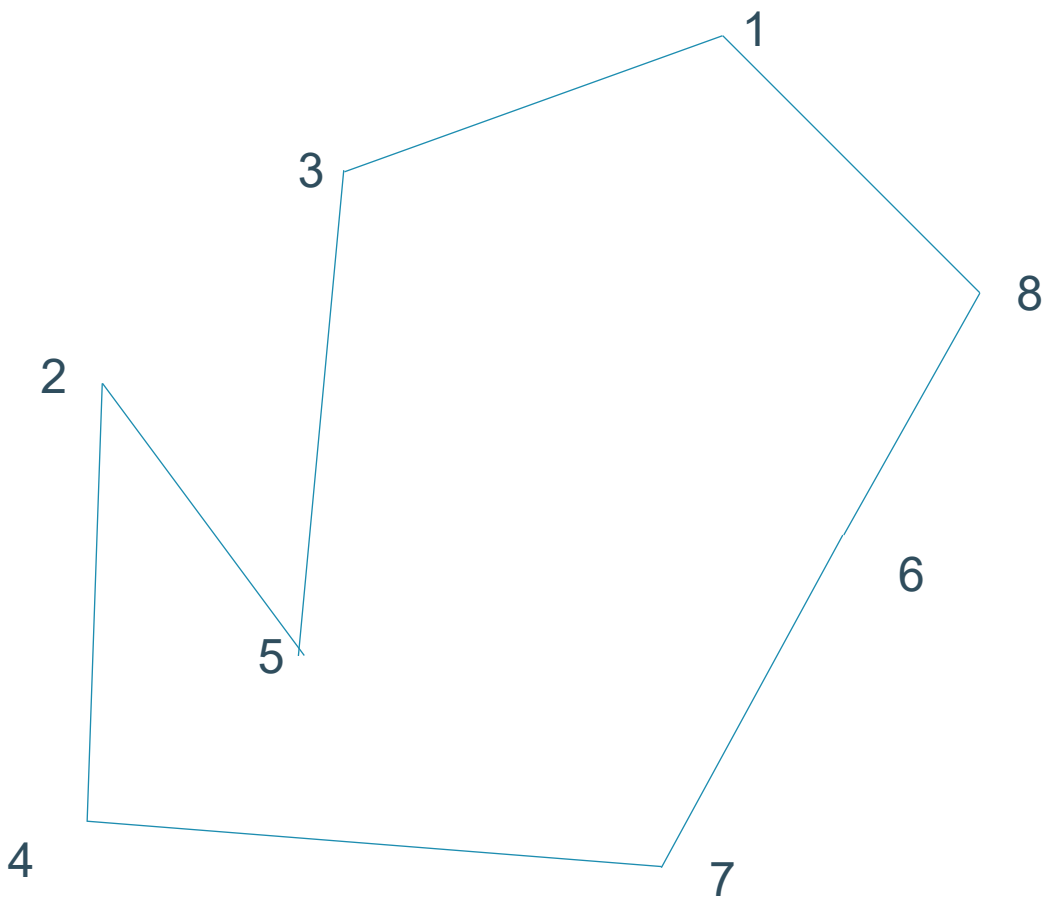
- Example 1: Traveling Salesman Problem (TSP)

- In an instance of the TSP we are given an integer n > 0 and the distance between every pair of n cities in the form of an n×n matrix [$d_{ij}$], where $d_{ij} \in Z^+$.
- A tour is a closed path that visits every city exactly once.
- The problem is to find a tour with minimal total length. F is then:
  - $F = \{\text{all cyclic permutations } \pi \text{ on } n \text{ objects}\}$
- If $\pi(j)$ is the city visited after city *j* for *j=1,…,n,* then *c* maps $\pi$ to:
  - $\sum_{j=1}^{n} d_{j\pi(j)}$

# Traveling salesman problem

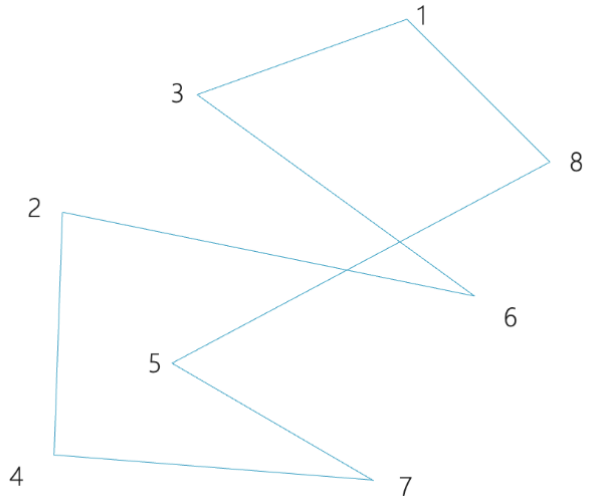


# Traveling salesman problem



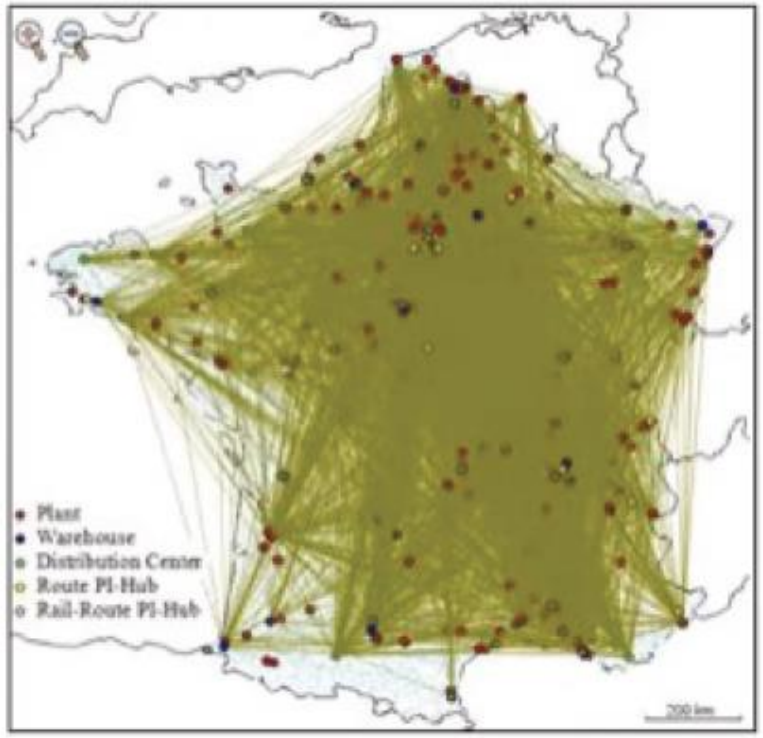
# Traveling salesman problem

- Solutions:
  - all possible permutations
  - 8 cities = 8! = 40320
  - 20 cities= 20! = 2,432902008176640e18



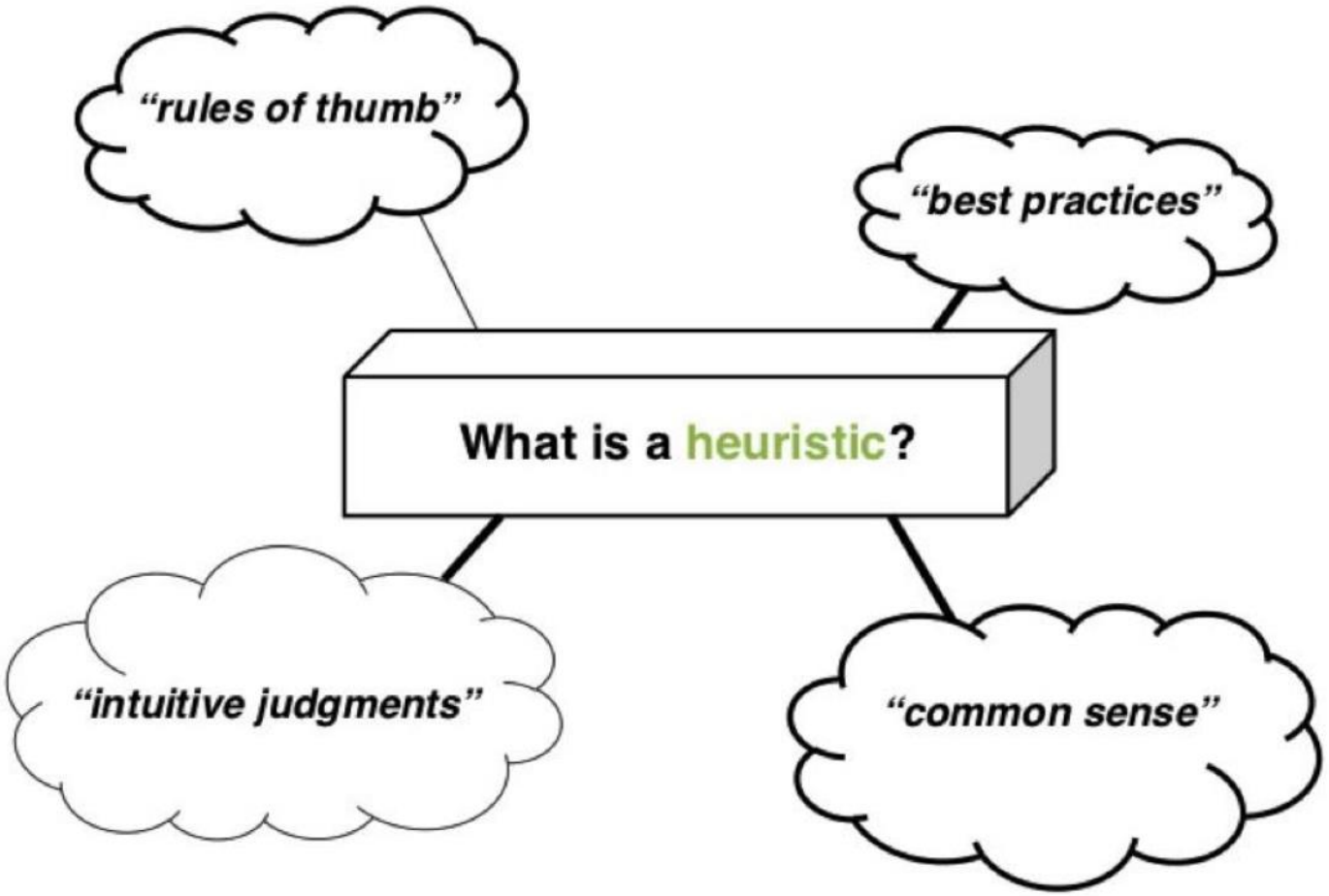
# Logistics network in France

- E. Ballot, B. Montreuil, R. Meller, The Physical Internet: The Networkof Logistics Networks, La documentation Francaise. Paris, France, 2014



# How to find a better solution?

- Use heuristics!



# Properties of heuristics

+ Fast
+ Scalable
+ High quality solutions
+ Solve any type of problem (complex constraints, non-linear, …)

- Problem specific (but usually transferable to other problems)
- Cannot guarantee optimality

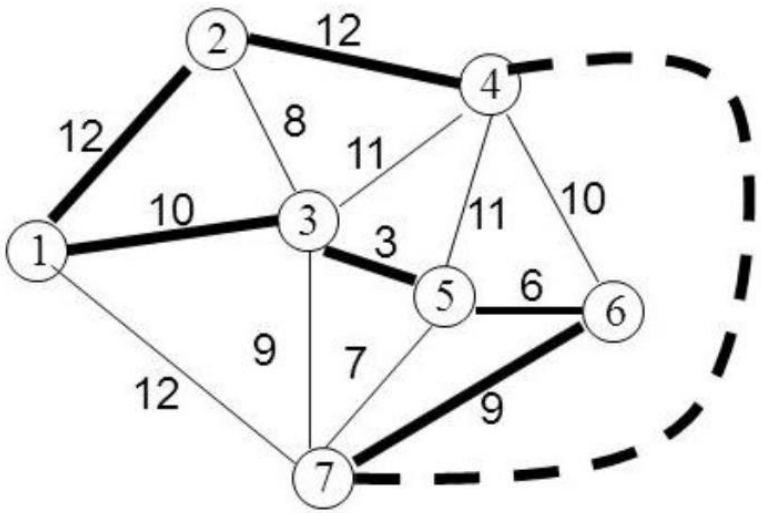
# Heuristic types

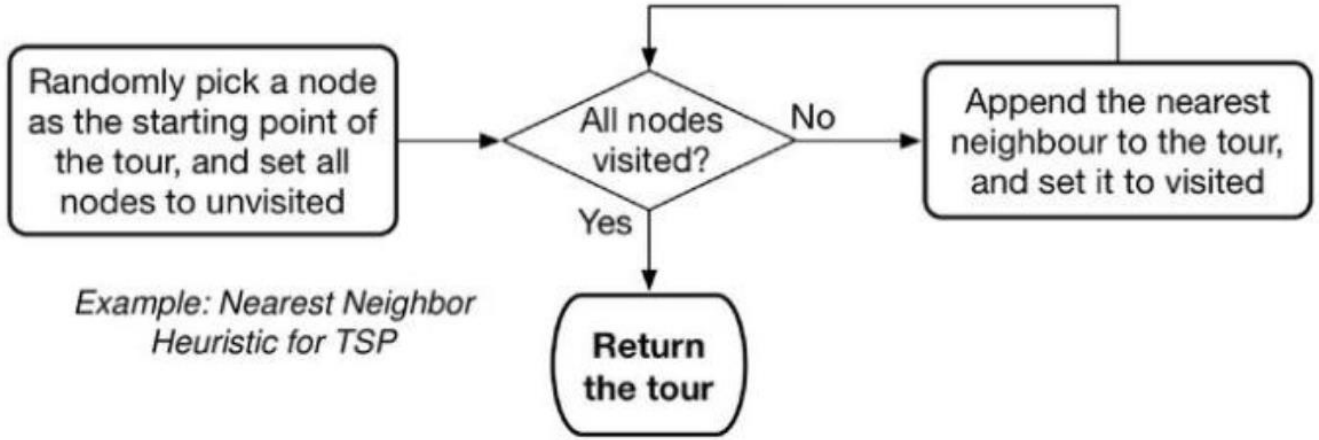
- **Constructive heuristics**
- **Metaheuristics**

- Hybrid heuristics
  - Matheuristics: hybrid of Metaheuristics and Mathematical Programming
  - Other hybrids: Machine Learning, Constraint Programming,…
- Hyperheuristics

# Constructive heuristics

- Incrementally construct a solution from scratch.
  - Easy to understand and implement
  - Usually very fast
  - Reasonably good solutions




*Example: Nearest Neighbor Heuristic for TSP*

# How to improve? Neighbourhoods!

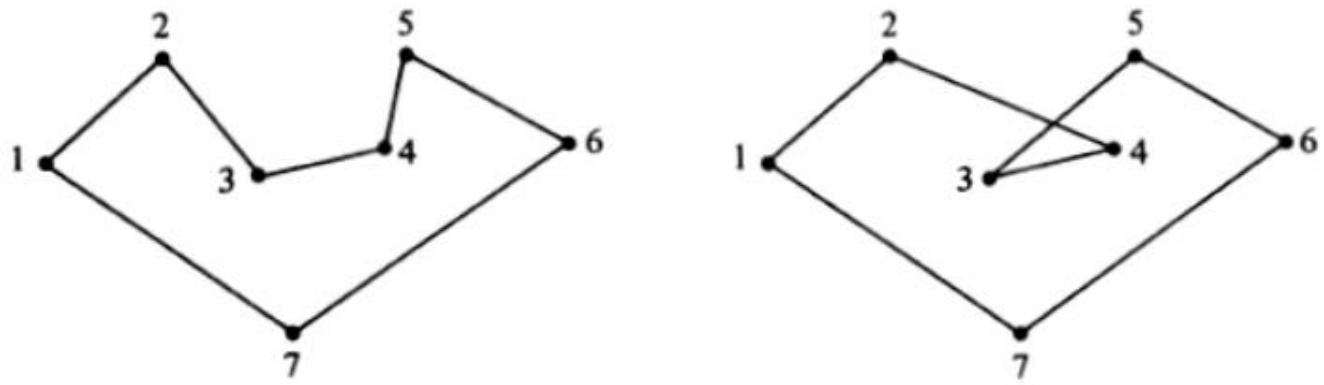
- Given a feasible point $\mathrm{f} \in \mathrm{F}$ in a particular problem, it is useful in many situations to define a set N(f) of points that are “close” in some sense to the point f.

- Definition:
  - Given an optimisation problem with instances (F, c), a neighbourhood is a mapping $N: F \longrightarrow 2^F$ defined for each instance.

- If $F = \mathbb{R}^n$, the set of points within a fixed Euclidean distance provides a natural neighborhood. In many combinatorial problems, the choice of N may depend critically on the structure of F.

# Neighbourhoods: TSP example

- 2-change neighbourhood for TSP




- $N_2(f) = \{g\colon g \in F \text{ and } g \text{ can be obtained from } f \text{ as follows}:$
  remove $two\ non-adjectent$ edges from the tour;
  then replace them with two edges$\}$
- Can be generalised to $N_k$

# Local and global optima

- Finding a globally optimal solution to an instance of some problems can be prohibitively difficult, but it is often possible to find a solution *f* which is best in the sense that there is nothing better in its neighbourhood *N(f)*.

- Definition:
  - Given an instance (F, c) of an optimisation problem and a neighbourhood N, a feasible solution $f \in F$ is called locally optimal with respect to N (or simply locally optimal whenever N is understood by context) if

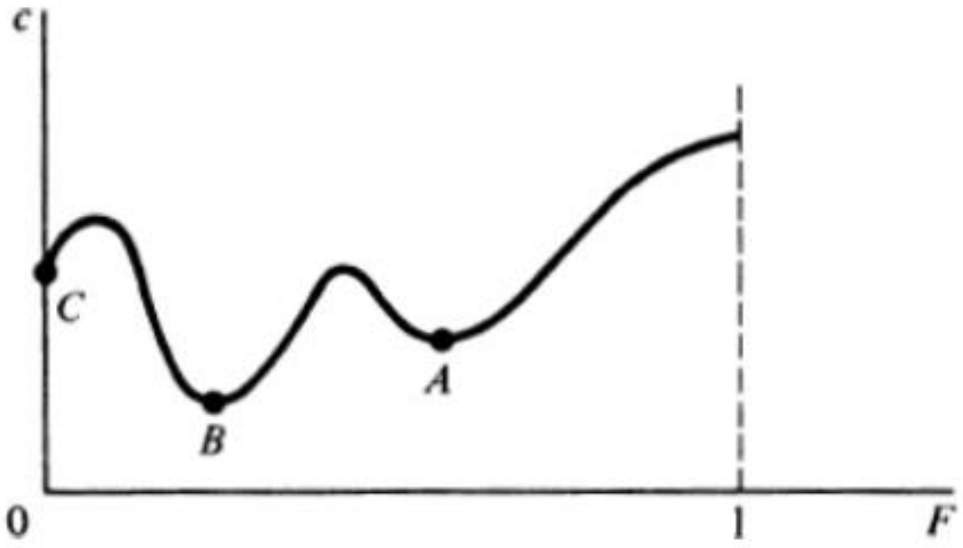
$$c(f) \leq c(g) \text{ for all } g \in N(f)$$

# Local and global optima

- Consider the instance of an optimisation problem (F,c) defined by

$$F = [0,1] \subseteq R$$

- and the cost function c shown below:



# Local, global optima, neighbourhoods and TSP

- In the TSP, solutions locally optimal with respect to the k-change neighborhood $N_k$ are called k-opt.

- To find a k-opt tour in an instance of the TSP use, for $t \in F$:

$$\text{improve}(t) = \begin{cases} \text{any } s \in N_k(t) \text{ such that } c(s) < c(t), \text{ if such an } s \text{ exists} \\ \text{'no' otherwise} \end{cases}$$

# Local, global optima, neighbourhoods and TSP

- First algorithm ‘to solve TSP’:

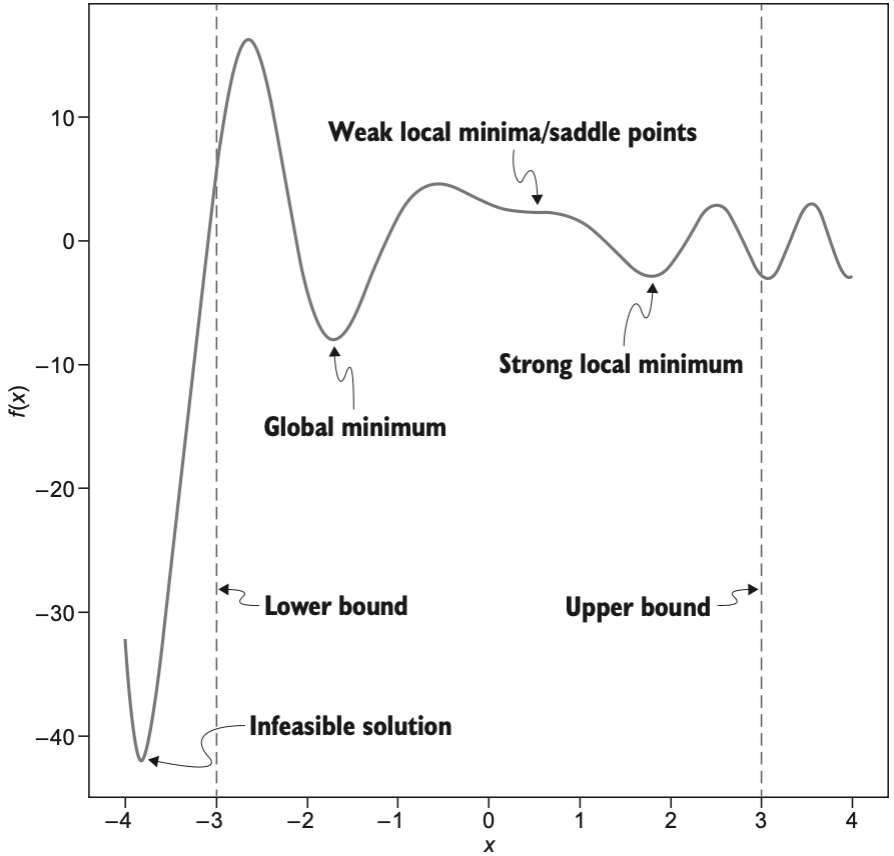
$$
\begin{array}{l}
\textbf{procedure } k\text{-opt} \\
\quad \textbf{begin} \\
\quad\quad t := \text{ some initial tour;} \\
\quad\quad \textbf{while } \text{improve}(t) \neq \text{‘no’ } \textbf{do} \\
\quad\quad\quad t := \text{improve}(t); \\
\quad\quad \textbf{return } t \\
\quad \textbf{end}
\end{array}
$$

- Optimal? Local? Global?

# More on solutions

- Solutions can be classified into:
  - Feasible: solutions that satisfy all the given constraints
  - Optimal: solutions that are both feasible and provide the best objective function value (cfr. cost)
  - Near-optimal: solutions that are feasible and provide a superior objective function value but are not necessarily the best

# More on solutions




A *global optimum* (or a *global minimum* in the case of minimization problems) is the best of a set of candidate solutions (i.e., the lowest point of the entire feasible search space). Mathematically, if $f(x)$ is the objective function, a point x* is the global minimum if, for all x in the domain of $f$, $f(x^*) \leq f(x)$.

A *strong local minimum* is a point where the function's value is less than (or equal to) the values of the function in a neighborhood around that point but is higher than the global minimum. Mathematically, a point x* is a strong local minimum if there is a neighborhood $N$ of $x^*$ such that $f(x^*) < f(x)$ for all $x$ in $N$ with $x \neq x^*$.

A *weak local minimum* is a point where the function's value is less than or equal to the function's values at neighboring points, but there are sequences of points converging to this point for which the function's values strictly decrease. Mathematically, a point $x^*$ is a weak local minimum if there is a neighborhood $N$ of $x^*$ such that $f(x^*) \leq f(x)$ for all $x$ in $N$.

# Constraints

- Constrained optimisation problems have a set of equality and/or inequality constraints $g_i(x)$, $h_j(x)$ that restrict the values assigned to the decision variables. In addition, most problems have a set of boundary constraints, which define the domain of values for each variable.
- Constraints can be hard (must be satisfied) or soft (desirable to satisfy).
- Examples from a school timetabling problem:
  - Not having multiple lectures in the same room at the same time is a hard constraint.
  - Not having a teacher give multiple lectures at the same time is also a hard constraint.
  - Guaranteeing a minimum of three teaching days for every teacher may be a soft constraint.
  - Locating back-to-back lectures in nearby rooms may be a soft constraint.
  - Avoiding scheduling very early or very late lectures may also be a soft constraint.

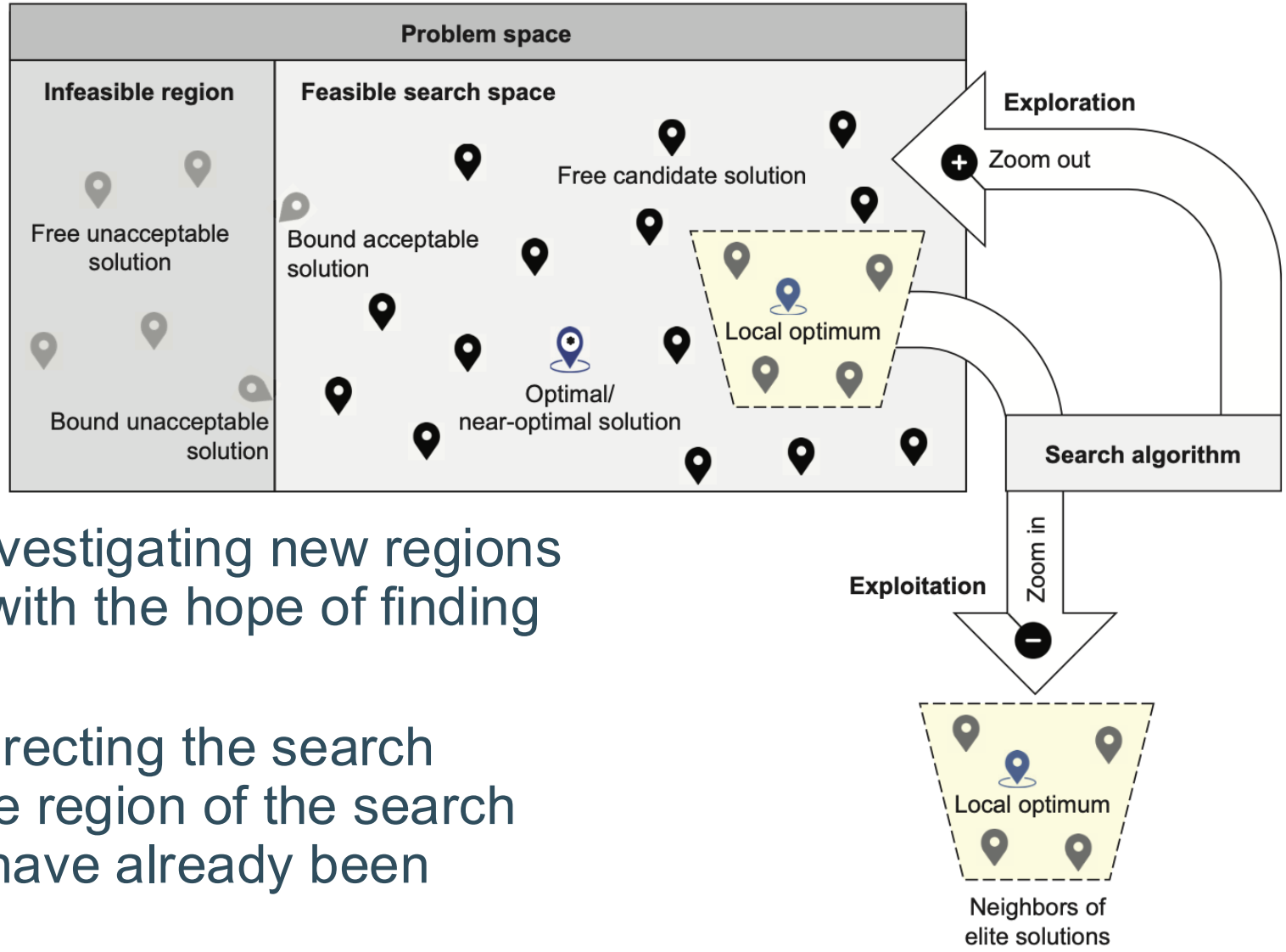
# Search space

- The goal of any optimisation method is to *assign values to decision variables* so that the *objective function* is optimised. To achieve this, *optimisation algorithms search the solution space* for candidate solutions. *Constraints* are simply *limitations on specific regions in the search space*. Thus, all such optimisation techniques can be considered **search methods**, where the goal is to find feasible solutions to satisfy constraints and maximize (or minimize) the objective functions.

# Search

- We'll define "search" as the *systematic examination of feasible states*, starting from the *initial state*, and ending (hopefully) at the *goal state*. However, while we explore the feasible search space, we may find a few reasonably good *neighbouring* solutions, and the question is whether we should *exploit* this region or keep *exploring*, looking for better solutions in other regions of the feasible search space.

# Search dilemma




- Exploration: the process of investigating new regions in the feasible search space with the hope of finding other promising solutions.
- Exploitation: the process of directing the search agent to focus on an attractive region of the search space where good solutions have already been found.

# Programming example – toy problem TSP

- Jupyter notebook.

# Metaheuristics

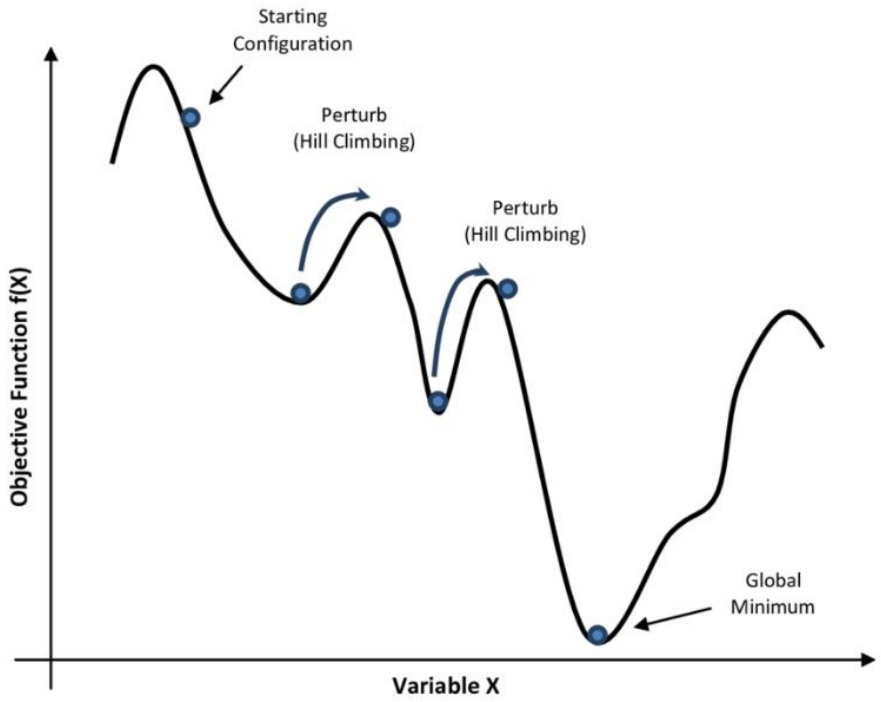
A **Metaheuristic** is a high-level problem independent algorithmic framework that provides a set of guidelines or strategies to develop heuristic optimization algorithms.

Instead of reinventing the wheel when developing a heuristic for each different problem, metaheuristics offer several tools that can be used as the skeleton for the development of a heuristic.

# Metaheuristics

Key property: iteratively improve one or more solutions

- Can be combined with constructive heuristics (initial solution)
- Local search based
  - Steepest descent / *Hill climbing*
  - Tabu search, Variable Neighbourhood Search (VNS)
  - *Simulated Annealing*, Late Acceptance
  - GRASP, Guided Local Search
- Population based
  - *Genetic algorithms / memetic algorithms*
  - Ant Colony Optimization
  - PSO (Particle swarm optimization)



INTERNATIONAL TRANSACTIONS IN OPERATIONAL RESEARCH

WILEY

INTERNATIONAL TRANSACTIONS IN OPERATIONAL RESEARCH

Intl. Trans. in Op. Res. 22 (2015) 3–18
DOI: 10.1111/itor.12001

# Metaheuristics—the metaphor exposed

Kenneth Sörensen

*University of Antwerp, Operations Research Group ANT/OR, Prinsstraat 13 – B5xx, 2000 Antwerp, Belgium*
*E-mail: kenneth.sorensen@ua.ac.be [Sörensen]*

Received 2 November 2012; received in revised form 12 November 2012; accepted 13 November 2012

**Abstract**

In recent years, the field of combinatorial optimization has witnessed a true tsunami of "novel" metaheuristic methods, most of them based on a metaphor of some natural or man-made process. The behavior of virtually any species of insects, the flow of water, musicians playing together – it seems that no idea is too far-fetched to serve as inspiration to launch yet another metaheuristic. In this paper, we will argue that this line of research is threatening to lead the area of metaheuristics away from scientific rigor. We will examine the historical context that gave rise to the increasing use of metaphors as inspiration and justification for the development of new methods, discuss the reasons for the vulnerability of the metaheuristics field to this line of research, and point out its fallacies. At the same time, truly innovative research of high quality is being performed as well. We conclude the paper by discussing some of the properties of this research and by pointing out some of the most promising research avenues for the field of metaheuristics.

# Metaheuristics - history

- Early research on heuristics focused on human intuition, insight and learning as a way of solving problems.

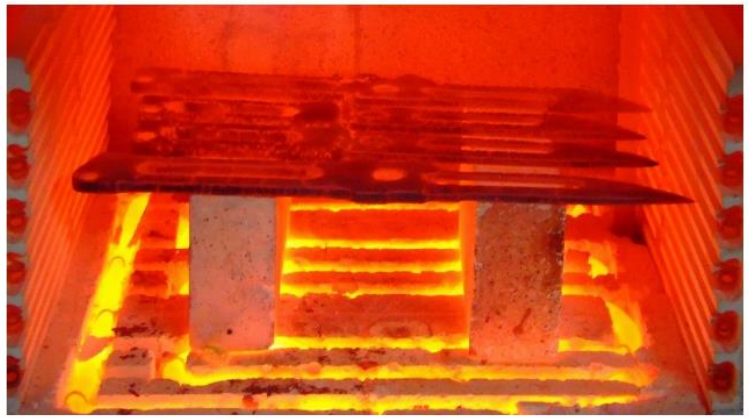
- Metaheuristics followed the same strategy.

- Example: Tabu search
  - Memorizes part of the trajectory the search had taken in the recent and not-so-recent past.
  - Similar to a person looking for water in the desert trying to avoid running in circles.

# Metaheuristics - history

- Eventually new methods were based on processes that had little to do with human intuition.
  - Simulated annealing
  - Natural evolution
  - Ant-colony optimization

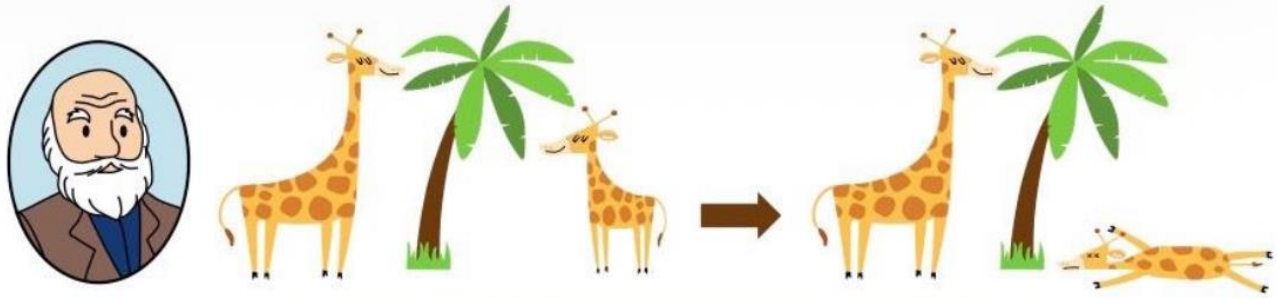
# History



- **Simulated annealing**
  - One of the earliest metaphor-based heuristics.
  - Analogy to the thermodynamic process of annealing.
    - Attempt to settle atoms into the lowest possible energy state.

  - Total energy state of the atoms ➡ quality of the solution.
  - Change of atom positions ➡ changes in the solution (moves).

# History



**Charles Darwin** – Evolution by *Descent with Modification* (1859)

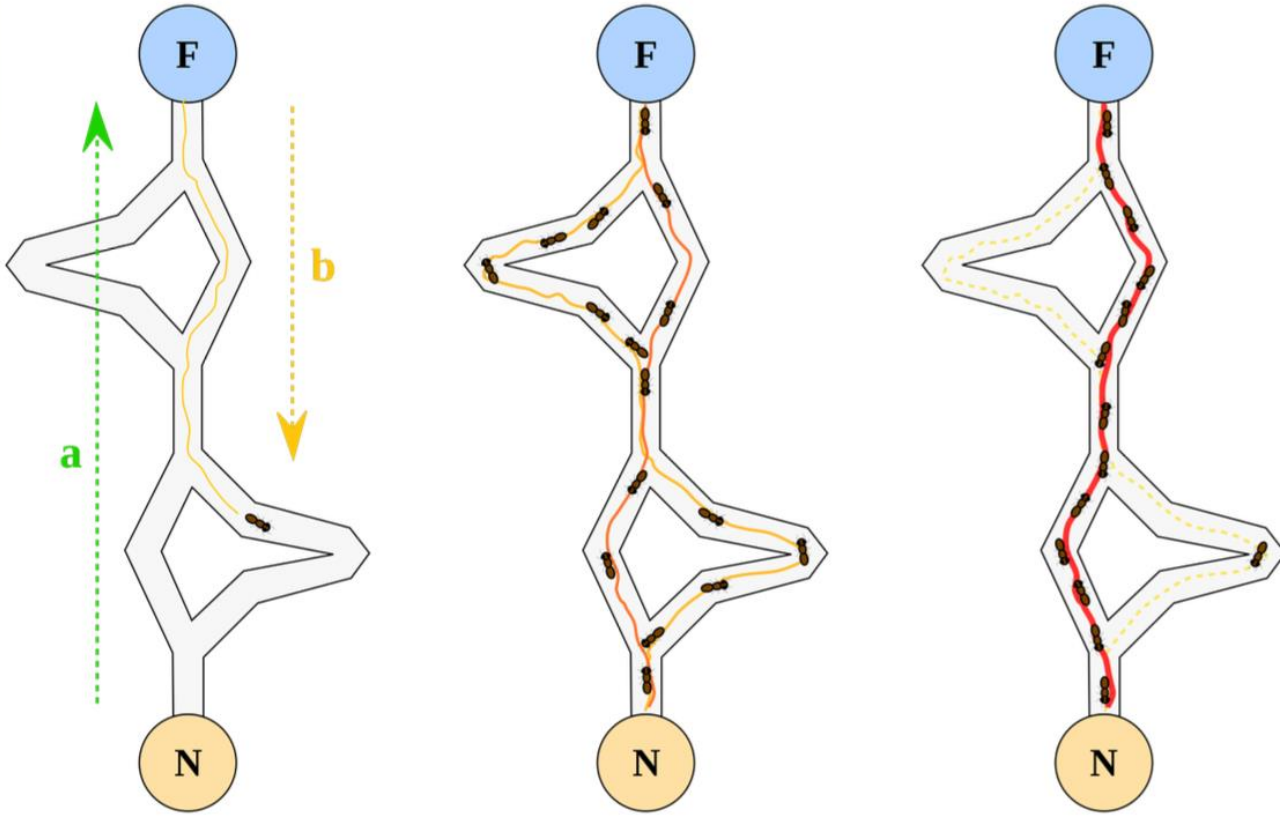
Long-necked giraffes are randomly born and have more offspring due to their competitive advantage

- **Natural evolution**
  - Probably the most successful metaphor used until now.
  - Individuals in a population competing for survival.

  - Individuals or chromosomes ➡ solutions.
  - Fitness ➡ objective function value.
  - Crossover / Mutation ➡ changes in the solution.

# History

- **Ant-colony optimization**
  - Models the optimization process upon the behavior of a colony of ants searching for food.
  - Agents (*ants*) construct solutions in parallel.
  - Each element of each solution has its own measure
    - *pheromone*
  - Elements with *stronger pheromones* have higher probability of being selected for future solutions.



# History

With the increase in number of metaheuristics being created comes the question:

*Until what point creating metaphors would result in really effective approaches?*

# Metaphors

- “Models of nature that are relied upon for inspiration [in creating new metaheuristics] are ubiquitous, and it is easy to conjure up examples whose metaphorical possibilities have not yet been tapped.[…]

- It is perhaps surprising in retrospect that the behavior of bees has not been selected as a basis for one of the ‘new’ problem solving methods.”
  – Glover & Laguna (1998)

# Metaphors

- People took seriously what he meant as sarcasm:
  - Bees optimization algorithm (2004).
  - Bee colony algorithm (2005).
  - Bee mating optimization algorithm (2006).

- ***Difference between all insect algorithms proved marginal at best.***

# Metaphors

Check for updates

Water Supply

IWA PUBLISHING

© 2022 The Authors

Water Supply Vol 22 No 4, 3702 doi: 10.2166/ws.2022.010

## A state-of-the-Art review of heuristic and metaheuristic optimization techniques for the management of water resources

Vijendra Kumar [a,*] and S. M. Yadav[b]

[a] Civil Engineering Department, G H Raisoni College of Engineering and Management, Pune, Maharashtra, India
[b] Civil Engineering Department, Sardar Vallabhbhai National Institute of Technology, Surat, Gujarat, India
*Corresponding author. E-mail: vijendra.kumar@raisoni.net, vij100000@gmail.com

VK, 0000-0002-0053-1210

### ABSTRACT

Water resource management is a complex engineering problem, due to the stochastic nature of inflow, various demands and environmental flow downstream. With the increase in water consumption for domestic use and irrigation, it becomes more challenging. Many more difficulties, such as non-convex, nonlinear, multi-objective, and discontinuous functions, exist in real life. From the past two decades, heuristic and metaheuristic optimization techniques have played a significant role in managing and providing better performance solutions. The popularity of heuristic and metaheuristic optimization techniques has increased among researchers due to their numerous benefits and possibilities. Researchers are attempting to develop more accurate and efficient models by incorporating novel methods and hybridizing existing ones. This paper's main contribution is to show the state-of-the-art of heuristic and metaheuristic optimization techniques in water resource management. The research provides a comprehensive overview of the various techniques within the context of a thorough evaluation and discussion. As a result, for water resource management problems, this study introduces the most promising evolutionary and swarm intelligence techniques. Hybridization, modifications, and algorithm variants are reported to be the most successful for improving optimization techniques. This survey can be used to aid hydrologists and scientists in deciding the proper optimization techniques.

**Key words:** evolutionary algorithm, optimization, reservoir operation, swarm intelligence, water management

### HIGHLIGHTS

- This paper reviews the different evolutionary algorithms and their applications.
- This paper reviews the different swarm intelligence algorithms and their applications.
- It identifies the importance and demerits of different heuristic and meta-heuristic algorithms.
- It highlights some of the observations on the challenges and needs.
- It highlights the state-of-the-art of water resources management and planning.

# Metaphors

Water Supply Vol 22 No 4, 3704

Check for updates

Water Supply

IWA PUBLISHING

© 2022 The Authors

Water Supply Vol 22 No 4, 3702 doi: 10.2166/ws.2022.010

**A state-of-the-Art review of heuristic and metaheuristic optimization techniques for the management of water resources**

Vijendra Kumar [a,*] and S. M. Yadav[b]

[a] Civil Engineering Department, G H Raisoni College of Engineering and Management, Pune, Maharashtra, India
[b] Civil Engineering Department, Sardar Vallabhbhai National Institute of Technology, Surat, Gujarat, India
*Corresponding author. E-mail: vijendra.kumar@raisoni.net, vij100000@gmail.com

VK, 0000-0002-0053-1210

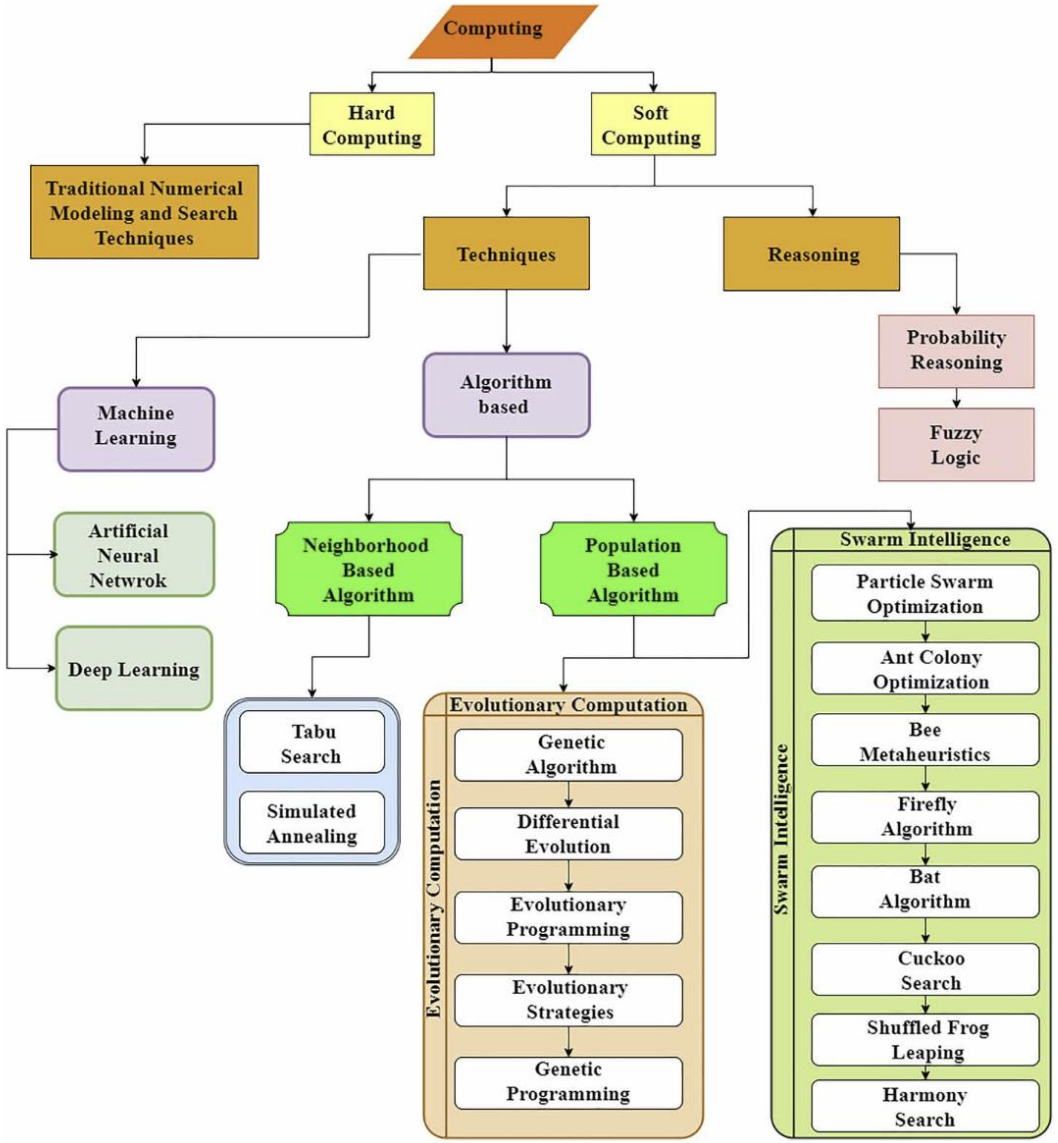
ABSTRACT

Water resource management is a complex engineering problem, due to the stochastic nature of inflow, various demands and environmental flow downstream. With the increase in water consumption for domestic use and irrigation, it becomes more challenging. Many more difficulties, such as non-convex, nonlinear, multi-objective, and discontinuous functions, exist in real life. From the past two decades, heuristic and metaheuristic optimization techniques have played a significant role in managing and providing better performance solutions. The popularity of heuristic and metaheuristic optimization techniques has increased among researchers due to their numerous benefits and possibilities. Researchers are attempting to develop more accurate and efficient models by incorporating novel methods and hybridizing existing ones. This paper's main contribution is to show the state-of-the-art of heuristic and metaheuristic optimization techniques in water resource management. The research provides a comprehensive overview of the various techniques within the context of a thorough evaluation and discussion. As a result, for water resource management problems, this study introduces the most promising evolutionary and swarm intelligence techniques. Hybridization, modifications, and algorithm variants are reported to be the most successful for improving optimization techniques. This survey can be used to aid hydrologists and scientists in deciding the proper optimization techniques.

**Key words:** evolutionary algorithm, optimization, reservoir operation, swarm intelligence, water management

HIGHLIGHTS

- This paper reviews the different evolutionary algorithms and their applications.
- This paper reviews the different swarm intelligence algorithms and their applications.
- It identifies the importance and demerits of different heuristic and meta-heuristic algorithms.
- It highlights some of the observations on the challenges and needs.
- It highlights the state-of-the-art of water resources management and planning.



# References


- Papadimitriou, C. H., & Steiglitz, K. (1998). Combinatorial optimization: Algorithms and complexity. Mineola, NY: Dover Publications.
- Khamis, A. (2024). Optimization Algorithms: AI techniques for design, planning, and control problems. Manning Publications.
- Introduction to metaheuristics. Elective course at KU Leuven, prof. Tony Wauters, Department of Computer Science.